

```

(venv) ser_123-holla@LAPTOP-6DQ3PM54:/mnt/c/Users/User/Desktop/smart_gym_and_fitness_project/Smart-Gym-Fitness-Center-Management-System/backend$ python run.py
[✓] Database connection pool created successfully

Starting Smart Gym API on http://localhost:5000
API Documentation: http://localhost:5000/api/docs (coming soon)
Health Check: http://localhost:5000/health
Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.14.182.194:5000
Press CTRL+C to quit
* Restarting with stat
[✓] Database connection pool created successfully

Starting Smart Gym API on http://localhost:5000
API Documentation: http://localhost:5000/api/docs (coming soon)
Health Check: http://localhost:5000/health
Auth Endpoints: http://localhost:5000/api/v1/auth

Press CTRL+C to stop the server

* Debugger is active!
* Debugger PIN: 111-100-485
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 404 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 401 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/payments/purchase-membership HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/payments/history HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/schedule HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/ HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "GET /api/v1/classes/schedule HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 401 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/auth/login HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 403 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 409 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 201 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 404 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "POST /api/v1/classes/book HTTP/1.1" 400 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "DELETE /api/v1/classes/cancel/54 HTTP/1.1" 200 -
127.0.0.1 - - [11/Mar/2026 01:18:32] "DELETE /api/v1/classes/cancel/54 HTTP/1.1" 400 -
]

```

TRANSACTIONS:

```
• (venv) ser_123-holla@LAPTOP-6DQ3PMS4:/mnt/c/Users/User/Desktop/smart_gym_and_fitness_project/Smart-Gym-Fitness-Center-
```

SMART GYM – STEP 4 TRANSACTION TESTS

Make sure you ran test_auth.py first to create test.member@gym.com
Make sure Flask is running (python run.py). Press Enter...

Auto-cleaning test data from previous runs...

Removed 2 test booking(s) for ahmed

Removed 1 test membership(s) for test.member

Removed 1 test payment(s) for test.member

Cleanup done – starting fresh!

```
=====
  SETUP: Getting tokens
=====
```

Member 1 (ahmed.khan) logged in

Member 2 (test.member) logged in

```
=====
  SCENARIO 1: MEMBERSHIP PURCHASE TRANSACTION
=====
```

Testing: Atomicity, Consistency, SELECT FOR UPDATE (locker), Durability

T01 | Missing tier_id → 400

Expected: 400 | Got: 400

{

"error": "tier_id is required",

"success": false

}

T02 | Invalid payment_method → 400

Expected: 400 | Got: 400

{

"error": "payment_method must be one of: ['cash', 'credit_card', 'debit_card', 'online']",

"success": false

}

T03 | Invalid tier_id=999 → 404 (transaction rolled back)

Expected: 404 | Got: 404

{

"error": "Membership tier 999 does not exist",

"success": false

}

T04 | No token → 401

Expected: 401 | Got: 401

{

"error": "Authorization token is missing",

"success": false

}

T05 | Member 1 already has membership → 409

Expected: 409 | Got: 409

{

"error": "You already have an active membership",

"existing_membership_id": 1,

"success": false

}

(venv) ser_123-h011a@LAPTOP-6DQ3PMS4:/mnt/C:/Users/User/Desktop/smart_gym_and_fitness_project/Sm

T06 | Member 2 (test.member) purchases Basic membership...

T06 | Purchase Basic membership → 201

Expected: 201 | Got: 201

```
{
  "message": "Membership purchased successfully",
  "success": true,
  "transaction": {
    "locker": null,
    "membership": {
      "end_date": "2026-04-11",
      "membership_id": 25,
      "start_date": "2026-03-11",
      "status": "active",
      "tier": "Basic"
    },
    "payment": {
      "amount": 5000.0,
      "method": "credit_card",
      "payment_date": "2026-03-11 01:18:32.483095",
      "payment_id": 27,
      "status": "completed"
    }
  }
}
```

Payment ID : 27

Membership : 25 (Basic)

Valid until : 2026-04-11

Locker : None (None for Basic – correct)

T07 | Duplicate membership → 409

Expected: 409 | Got: 409

```
{
  "error": "You already have an active membership",
  "existing_membership_id": 25,
  "success": false
}
```

T08 | Payment history → 200

Expected: 200 | Got: 200

```
{
  "count": 2,
  "payments": [
    {
      "amount": "2000.00",
      "description": "1-hour PT with Coach Ali",
      "payment_date": "Tue, 10 Feb 2026 16:00:00 GMT",
      "payment_id": 9,
      "payment_method": "credit_card",
      "payment_type": "training_session",
      "status": "completed"
    },
    {
      "amount": "8000.00",
      "description": "Premium membership - Jan 2026",
      "payment_date": "Thu, 01 Jan 2026 10:00:00 GMT",
      "payment_id": 1,
      "payment_method": "credit_card",
      "payment_type": "membership",
      "status": "completed"
    }
  ]
}
```

```
    ],  
    "success": true  
}
```

=====

SCENARIO 2: CLASS BOOKING TRANSACTION

=====

Testing: SELECT FOR UPDATE, trigger integration, waitlist logic

Fetching class schedules...

Found 4 upcoming schedules

Open: 3 | Full: 1

→ ID:16 Morning Yoga 2026-03-13 (15/15 spots)

→ ID:17 HIIT Blast 2026-03-26 (20/20 spots)

→ ID:18 Zumba Dance 2026-04-12 (25/25 spots)

T09 | Get classes list → 200

Expected: 200 | Got: 200

```
{  
  "classes": [  
    {  
      "class_id": 6,  
      "class_name": "Beginner Strength",  
      "class_type": "Strength",  
      "description": "Introduction to weight training",  
      "difficulty_level": "beginner",  
      "duration_minutes": 60,  
      "max_capacity": 10  
    },  
    {  
      "class_id": 4,  
      "class_name": "CrossFit WOD",  
      "class_type": "Strength",  
      "description": "Workout of the Day - functional fitness",  
      "difficulty_level": "advanced",  
      "duration_minutes": 60,  
      "max_capacity": 12  
    },  
    {  
      "class_id": 2,  
      "class_name": "HIIT Blast",  
      "class_type": "Cardio",  
      "description": "High-intensity interval training for fat loss",  
      "difficulty_level": "advanced",  
      "duration_minutes": 45,  
      "max_capacity": 20  
    },  
    {  
      "class_id": 1,  
      "class_name": "Morning Yoga",  
      "class_type": "Mind & Body",  
      "description": "Relaxing yoga session to start your day",  
      "difficulty_level": "beginner",  
      "duration_minutes": 60,  
      "max_capacity": 15  
    },  
    {  
      "class_id": 5,  
      "class_name": "Spin Cycle",  
      "class_type": "Cardio",  
      "description": "Indoor cycling with motivating music",  
      "max_capacity": 25  
    }  
  ]  
}
```

```

    },
    {
      "class_id": 5,
      "class_name": "Spin Cycle",
      "class_type": "Cardio",
      "description": "Indoor cycling with motivating music",
      "difficulty_level": "intermediate",
      "duration_minutes": 45,
      "max_capacity": 18
    },
    {
      "class_id": 3,
      "class_name": "Zumba Dance",
      "class_type": "Cardio",
      "description": "Fun Latin-inspired dance workout",
      "difficulty_level": "intermediate",
      "duration_minutes": 50,
      "max_capacity": 25
    }
  ],
  "count": 6,
  "success": true
}

```

T10 | Get schedule → 200

Expected: 200 | Got: 200

```

{
  "count": 4,
  "schedules": [
    {
      "class_name": "Morning Yoga",
      "class_type": "Mind & Body",
      "current_capacity": 0,
      "difficulty_level": "beginner",
      "duration_minutes": 60,
      "end_time": "07:00:00",
      "max_capacity": 15,
      "room_number": "Studio A",
      "schedule_date": "2026-03-13",
      "schedule_id": 16,
      "spots_available": 15,
      "start_time": "06:00:00",
      "status": "scheduled",
      "trainer_name": "Sara Nadeem"
    },
    {
      "class_name": "HIIT Blast",
      "class_type": "Cardio",
      "current_capacity": 0,
      "difficulty_level": "advanced",
      "duration_minutes": 45,
      "end_time": "18:45:00",
      "max_capacity": 20,
      "room_number": "Studio B",
      "schedule_date": "2026-03-26",
      "schedule_id": 17,
      "spots_available": 20,
      "start_time": "18:00:00",
      "status": "scheduled",
      "trainer_name": "Nida Khan"
    }
  ],
}

```

```

    },
    {
      "class_name": "Zumba Dance",
      "class_type": "Cardio",
      "current_capacity": 0,
      "difficulty_level": "intermediate",
      "duration_minutes": 50,
      "end_time": "19:50:00",
      "max_capacity": 25,
      "room_number": "Studio A",
      "schedule_date": "2026-04-12",
      "schedule_id": 18,
      "spots_available": 25,
      "start_time": "19:00:00",
      "status": "scheduled",
      "trainer_name": "Sara Nadeem"
    },
    {
      "class_name": "Beginner Strength",
      "class_type": "Strength",
      "current_capacity": 10,
      "difficulty_level": "beginner",
      "duration_minutes": 60,
      "end_time": "11:00:00",
      "max_capacity": 10,
      "room_number": "Studio C",
      "schedule_date": "2026-05-13",
      "schedule_id": 19,
      "spots_available": 0,
      "start_time": "10:00:00",
      "status": "scheduled",
      "trainer_name": "Ali Rehman"
    }
  ],
  "success": true
}

```

T11 | Book without token → 401

Expected: 401 | Got: 401

```

{
  "error": "Authorization token is missing",
  "success": false
}

```

T12 | Non-member role → 403

Expected: 403 | Got: 403

```

{
  "error": "Access denied. Required role(s): m, e, m, b, e, r",
  "success": false,
  "your_role": "admin"
}

```

T13 | Booking: Morning Yoga on 2026-03-13 (15 spots)

T13 | Confirmed booking → 201

Expected: 201 | Got: 201

```

{
  "booking": {
    "booking_id": 54,
    "class_name": "Morning Yoga",
    "position_in_waitlist": null,

```

```
T13 | Confirmed booking → 201
  Expected: 201 | Got: 201
  {
    "booking": {
      "booking_id": 54,
      "class_name": "Morning Yoga",
      "position_in_waitlist": null,
      "schedule_date": "2026-03-13",
      "spots_remaining": 14,
      "start_time": "06:00:00",
      "status": "confirmed"
    },
    "message": "Class booked successfully \u2014 confirmed!",
    "success": true
  }
  Booking ID : 54
  Status      : confirmed
  Spots left   : 14

T14 | Duplicate booking → 409
  Expected: 409 | Got: 409
  {
    "error": "You have already booked this class",
    "existing_status": "confirmed",
    "success": false
  }

T15 | Full class: Beginner Strength (0 spots)

T15 | Full class → waitlist → 201
  Expected: 201 | Got: 201
  {
    "booking": {
      "booking_id": 55,
      "class_name": "Beginner Strength",
      "position_in_waitlist": 1,
      "schedule_date": "2026-05-13",
      "start_time": "10:00:00",
      "status": "waitlist"
    },
    "message": "Class is full. Added to waitlist at position 1.",
    "success": true
  }
  Waitlist position: 1

T16 | Invalid schedule_id=99999 → 404
  Expected: 404 | Got: 404
  {
    "error": "Schedule 99999 not found",
    "success": false
  }

T17 | Missing schedule_id → 400
  Expected: 400 | Got: 400
  {
    "error": "No data provided",
```

```

T17 | Missing schedule_id → 400
    Expected: 400 | Got: 400
    {
      "error": "No data provided",
      "success": false
    }

T18 | Cancel booking → 200
    Expected: 200 | Got: 200
    {
      "cancelled_booking_id": 54,
      "message": "Booking cancelled successfully",
      "note": "If confirmed, first waitlist member was automatically promoted",
      "success": true
    }

T19 | Cancel already cancelled → 400
    Expected: 400 | Got: 400
    {
      "error": "Booking is already cancelled",
      "success": false
    }

```

```

=====
EXPECTED RESULTS SUMMARY
=====

```

```

SCENARIO 1 - Membership Purchase:

```

```

T01: 400 – missing tier_id (validation)
T02: 400 – invalid payment method (validation)
T03: 404 – tier not found (transaction rolled back)
T04: 401 – no token (auth)
T05: 409 – already has membership (consistency)
T06: 201 – membership purchased (ACID commit)
T07: 409 – duplicate blocked (consistency)
T08: 200 – payment history

```

```

SCENARIO 2 - Class Booking:

```

```

T09: 200 – public classes list
T10: 200 – public schedule
T11: 401 – no token
T12: 403 – wrong role
T13: 201 – confirmed booking (SELECT FOR UPDATE)
T14: 409 – duplicate booking blocked
T15: 201 – waitlist position (class full)
T16: 404 – invalid schedule
T17: 400 – missing schedule_id
T18: 200 – cancellation (trigger promotes waitlist)
T19: 400 – already cancelled

```